# Charan Vardham

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## Summary

Computer Science undergraduate (expected graduation April 2028) seeking a Machine Learning and AI Engineer Intern role. Hands on experience designing, training, evaluating, and deploying ML models end to end, including deep learning in PyTorch for medical imaging (research submitted to SPIE Medical Imaging 2027), GPT2 pretraining on H100 GPUs with a custom FlashAttention2 kernel, and a fraud detection ensemble on AWS. Built LLM applications and AI agents with RAG and LangGraph, plus PyTorch open source contributions.

## Education

**University of Wisconsin Madison** — Madison, WI

*Bachelor of Science in Computer Science, Minor in Mathematics, 3.9 GPA (CWRU transfer)* — *Aug 2024 to April 2028*

- **Relevant Courses:** Data Structures, Algorithms, Discrete Mathematics, Linear Algebra, Computer Security

## Experience

**AI/ML Intern** — May 2025 to Aug 2025

*IQVIA* — *Remote*

- Built a **RAG** chatbot using **LLMs** over **200MB+** of clinical documents, achieving **92% Recall@5** on 200 labeled queries
- Integrated NVIDIA Riva APIs for real time STT/TTS with an LLM backend in a voice enabled chatbot

**Undergraduate Student Researcher** — Aug 2025 to Aug 2026

*Case Western Reserve University* — *Cleveland, OH*

- Trained deep learning CNN models in PyTorch on **5000+** CT slices for image reconstruction, evaluated with PSNR and SSIM
- Outperformed Pix2Pix, CycleGAN, and CUT baselines by up to **54%** lower FID; submitted to SPIE Medical Imaging 2027
- Ran **DDP** training with torchrun on multi GPU SLURM nodes, debugging NCCL timeouts and out of memory crashes
- Processed DICOM data with pydicom and built KernelEstimator and FilterEstimator models for CT kernel conversion

**Open Source Contributor** — May 2026 to Present

*PyTorch* — *Remote*

- Fixed **3** bugs in PyTorch Inductor's `torch.compile()`, achieving a **2x** improvement in graph compilation speed
- Added CPython's `descr_check` validation to Dynamo's C descriptor variable tracking, fixing incorrect tracing control flow

**xLab Intern** — Jan 2026 to May 2026

*Weatherhead School of Management, CWRU* — *Cleveland, OH*

- Developed an AI powered case study platform using **LLMs** to simulate real world business decision making
- Built the full stack application with **Next.js** (**TypeScript**), Tailwind CSS, **PostgreSQL**, and Prisma

**Undergraduate Student Researcher** — Sep 2026 to Present

*University of Wisconsin Madison* — *Madison, WI*

- Implementing a differential static timing analysis engine in **C++** with **reverse mode automatic differentiation**

## Projects

**GPT2 Pretraining** (link) | *PyTorch, Python, CUDA, Triton, SLURM* — February 2026

- Trained GPT2 (**124M**) from scratch on **10B** tokens on H100 GPUs at **260K** tokens/sec, reaching loss **3.0**
- Implemented a custom **FlashAttention2** kernel in **Triton**, improving throughput by **10 to 50%**

**AI Incident Response Platform** (link) | *LangGraph.js, TypeScript, Kafka, Inngest, Qdrant, Redis* — Jan 2026

- Built an event driven Kafka pipeline routing alerts to Inngest orchestrated LLM agents, cutting resolution to **15 minutes**
- Built agents that triage severity and investigate logs, metrics, and deploy history in parallel
- Designed a remediation agent that auto applies fixes or escalates to a human via Slack

**Real Time Credit Card Fraud Detection** (link) | *Python, XGBoost, AWS, Docker* — June 2026

- Built an end to end ML pipeline with a stacking ensemble (XGBoost, Random Forest, Logistic Regression) and threshold tuning
- Deployed a Dockerized FastAPI service on AWS ECS Fargate with DynamoDB, GitHub Actions CI/CD, and drift detection
- Implemented a Next.js inference API (predict, transaction history, health check) writing to DynamoDB

## Technical Skills

**Languages**: Python, SQL, C++, Java, TypeScript, Rust, Go, C#, Bash, R
**Generative AI / LLMs**: LLMs, AI agents, Agentic workflows (LangGraph), RAG pipelines, Prompt engineering, Vector databases (Qdrant), Hugging Face, Natural language processing, Retrieval evaluation, Multi agent systems
**Deep Learning**: PyTorch, TensorFlow, Transformer models, CNNs, GANs, Neural networks, Model training, Distributed training (DDP, NCCL), CUDA, Triton, FlashAttention, `torch.compile`, Automatic differentiation
**ML / Data**: Machine learning, Scikit Learn, XGBoost, Model evaluation, Feature engineering, Data pipelines, Pandas, NumPy
**MLOps / Tools**: Model deployment, Drift monitoring (Evidently AI), Docker, Kubernetes, AWS, FastAPI, Git, CI/CD